GeO$_2$

Inspiring Conscious Commuting

Group 3

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Project introduction

The problem, the brief and project background

The problem

The effects of air pollution can be drastic, ranging from coronary heart disease, stroke, lung cancer to other health conditions, of which there could be 2.5 million new cases by 2035 in the UK (Enekel, 2020). However, it is often overlooked, especially during commutes. From our desk research, we discovered that an estimated 3,799 deaths were caused in London by PM2.5 (particulate air pollution) alone in 2017 (Defra, 2019). How could help consumers and battle this issue through product design?

"an estimated 3,799 deaths were caused in London by PM2.5 (particulate air pollution) alone in 2017" (Defra, 2019)

Who?
All commuters

Where?
In London

When?
When commuting

How?
Ideate select and develop 3 concepts

The brief

An aspect that contributes to the issue is that air pollution is invisible- this means that it is often overlooked. As a result, the product will have to have a behavioural dimension to it. This is because in order to protect users from air pollution, we will have to make them care about doing so.

We want to innovate ways to help commuters better understand their exposure to air pollution, so that they might adapt their commute and reduce their risk of developing negative health effects.

HMW: How might we encourage users to alter their commutes to reduce their exposure to air pollution?

Our process

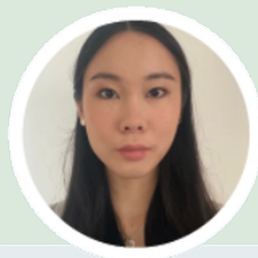
1. Defining the Problem
2. User research
3. Concept and feedback
4. Branding
5. Market analysis
6. CAD design
7. Pot prototypes
8. App UI prototype
9. Final concept and validation

User research

Research we conducted to further shed light on possible opportunities and product requirements

User research

*"I don't like the smell of air pollution, but I'd need **another incentive** to make me **change** my commute"*
- Juliette, student, 17



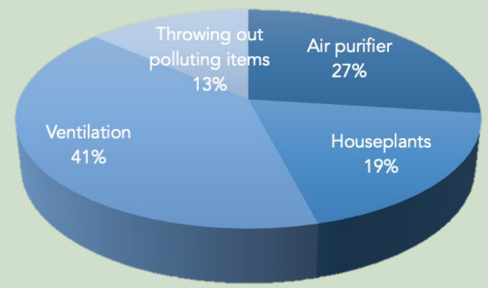
*"As a medicine student, I know about the **effects** of air pollution, although I can often **forget**"*
- Sophie, student, 19

*"Honestly, I **don't think** about it at all. I've never really **learned** that it can be bad for my health."*
- Lawrence, student, 13

*"I know air pollution can affect my health but **it isn't easy figuring out** how to **avoid** it during my commute"*
- Johan, banker, 52

Opportunities

We sent out a survey asking how people attempted at reducing air pollution in their homes, to try to form an understanding of what users were already familiar with. We noticed that **ventilation, air purifiers and houseplants** were the **top 3** choices. **Combining** two or more of those in could be a potential **opportunity** for our product.



Stakeholders

We spoke to a team of researchers whose focus was mainly on air pollution, who told us that there are currently two ways in which the problem of air pollution exposure is being tackled: either by purifying air, or creating a barrier between the human and the polluted air. In other words, there is the physical approach, and the behavioural one. Both approaches are rarely combined, and we saw this as an opportunity to look more into.

*"there's the **physical** approach of purifying air, and the **behavioural** one of protecting people from the remaining pollution. They're rarely combined, although this would be more effective"*

Requirements

From our user research, we drew up some requirements we wanted our product to fulfil:

- The product should give users a **strong incentive** to avoid air pollution
- The product should reinforce the user's **understanding** of the effects of air pollution, perhaps by visualizing it
- The product should **facilitate** the process of avoiding air pollution, so that users don't have to work about the 'how'

Selection of concept

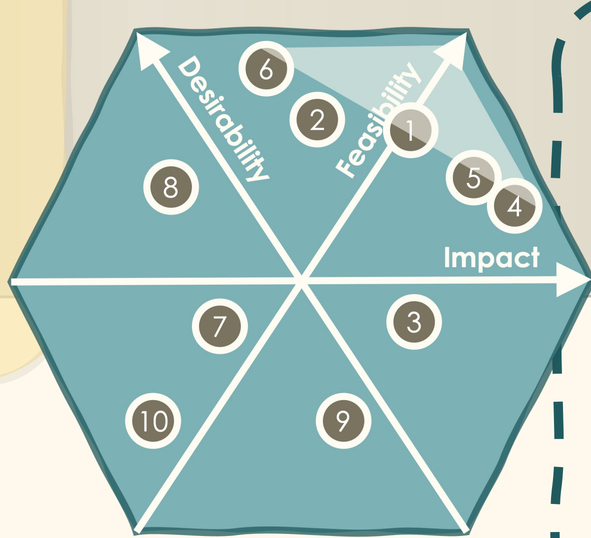
Narrowing down our 10 concepts that fit the brief, to a final one, using a range of selection techniques

Collation

First round of selection:

We started off by collating 10 of our individual ideas and plotting them according to desirability, feasibility and impact.

Radar graph



1- air purifying titanium dioxide plant pot
2- air purifying speaker
3- helmet- integrated air purifying mask
4- charcoal tube seats
5- bike display
6- virtual geocaches
7- air purifying clothing
8- street light indicators
9- digital personal assistant
10- air purifying furniture

SELECTION

We started asking the question ***"where and when does air actually need to be purified?"****, as this correlated with both the product's impact and desirability*

Second round of selection:

The 3 final concepts we could explore were:

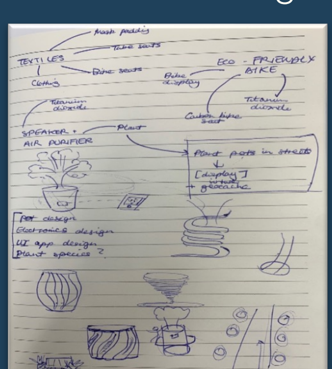
1- Eco-friendly bike (titanium dioxide coating, air pollution bike display, carbon padded seat)
2- Textiles and air purification (clothing, tube seats, helmet padding etc.)
3- Air-purifying plant pots in streets

SELECTION

Selection

We considered each of our 3 preliminary concepts from both qualitative and quantitative points of view. After team discussion followed by the completion of a matrix.

We settled on the **air purifying plant pot**. Some of the reasons why are detailed in the diagram on the right.



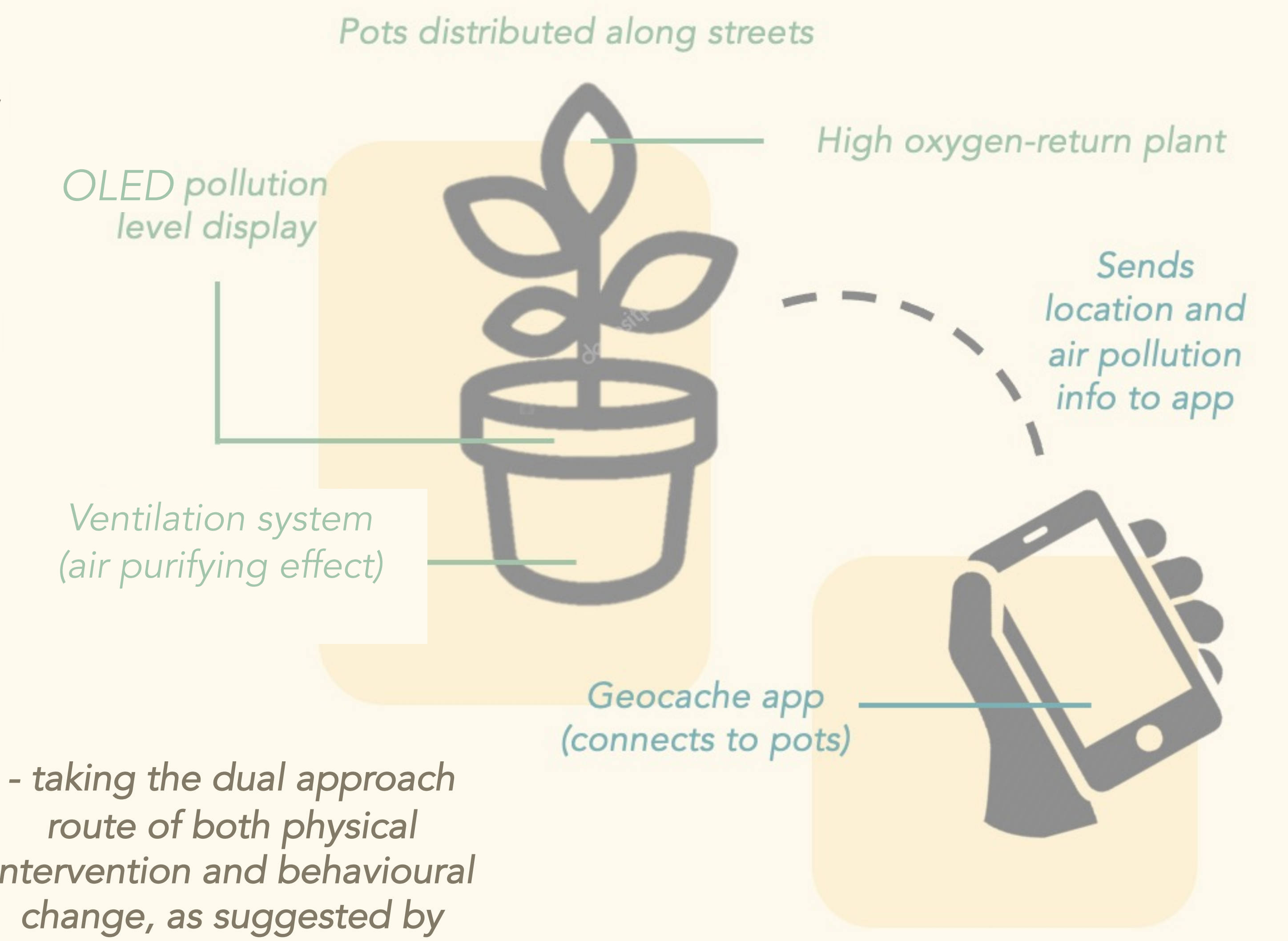
Concept/criteria	Effectiveness	Attractiveness	Eco-friendliness	Relevance	Sustainability	Feasibility	**Total**
1	3	5	5	5	4	2	24/30
2	1	2	1	5	3	3	15/30
3	4	5	4	5	4	4	26/30

Chosen concept and feedback

Defining our chosen concept, collecting user feedback and making conceptual changes accordingly

The Geocache, air-purifying plant pot

The plant pots are distributed along streets, and send air pollution data that they have sensed to the geocache app via Bluetooth. The app uses the data to encourage users to take 'cleaner' commutes with less air pollution, as well as to keep track of the user's exposure. Following our user feedback (see below), we added a gamification feature as an extra incentive. Users are further motivated to take alternative journeys as the app treats the plant pots as 'geocaches' which must be found.



- taking the dual approach route of both physical intervention and behavioural change, as suggested by stakeholders

- combining the two of the top 3 familiar methods against air pollution as discovered by our research (page 3): plants and air purifiers

Concept feedback

We then introduced our concept to 8 people, and asked them to rank their answers to the following questions on a scale of 1-10:

On a scale of 1-10...

how desirable do you find the product?

how rewarding would you expect it to be?

how likely is it that this would change your commuting habits?

Following the feedback, we realised that we had not completely fulfilled one of our requirements, as the incentive for taking a cleaner path was still not strong enough. Therefore, we decided to combine the air-purifying plant with another concept: the **virtual geocaches**. by bringing **gamification** to our product, we hoped to add an **excitement factor** that would strongly encourage users to try the product.

Branding

Establishing the visual identity, vision, essence and story of our brand

Visual identity

Brand name

GeO_2 is made up of the prefix 'geo' (**Earth**) that also alludes to **geocaching**, and the chemical symbol for **oxygen**, O_2, which relates to air purification.

Logo

We incorporated the compass to refer to geocaching, and finally chose a sleeker, minimalistic design.

Aesthetics

Natural/ earthly tones, **playful** forms, lightness/ airiness, **friendly** fonts, casual designs.

Brand vision

Vision

A **cleaner** city whose inhabitants **enjoy** taking less polluted commutes.

Differences

We combine **air purification** with mapping and **geocaching**, which hasn't been done before.

Objectives

To inspire **awareness** and **enthusiasm** in London citizens to take the **now-cleaner** path.

Brand Essence

Brand personality

A **cleaner** city whose inhabitants **enjoy** taking less polluted commutes.

Values

Our primary value is **health**, but why not encourage users to look after themselves whilst having **fun**?

Key words

Playfulness, **adventure**, **youthfulness**, **nature**, user-friendly, light-heartedness, casual.

Who we are

Our origins

We noticed that London citizens did little to **reduce**/ **avoid** air pollution, something we knew had to **change**.

Context

London is the most polluted city in the UK, with particulate pollution causing **3,799 deaths** in 2017.

Perspective

Our team combines two perspectives- changing **behaviour** and directly cleaning air.

Benchmarking and market analysis

Notable examples from our market research, corresponding insights we drew from them and the features and validation metrics we concluded our product should have.

Plant pot- AIRY

Features

Ventilation system creates air flow to roots so that plants can absorb up to **8x** as many pollutants as in conventional pots. Slick, clean design.

Pros + cons

+ No **electricity**
+ No replacement filters
+ **Cheap** to manufacture
- **Extra space** needed for ventilation

User responses

- Users don't **trust** that pot is actually effective
- Popular **aesthetic**
- Ventilation **compromises** normal functions- users found leaks and mould

Insights- our product

1) Biggest issue was that purifying effects are invisible… **find a way to visualise them**
2) Functional aspect **concealed in sleek design** is popular

App- Pokémon GO

Features

AR geocache app that encourages users to **travel** to find collectable, virtual creatures in real-life locations.

Pros + cons

+ Increased users' **fitness**
+ Gamification gives **excitement** factor
- Not **accessible** to all
- Isn't combined with **mapping**

User response

- Encouraged users to **explore** their area
- Gamification + animations were sometimes **excessive**

Insights- our product

1) Add a **mapping feature** to our app, as suggested by users in Pokémon GO
2) Gamification should **enhance** experience, but not dominate and hinder it

Validation metrics

Following our market research, we determined some qualities that our product should have:

- **Efficient**
As users will be using the product when they commute, the fun gamified aspect should not take away from the product's role as an alternative commute planner

- **User-friendly**
People from a range of ages, abilities and backgrounds use mapping apps, therefore, our product should be made accordingly.

- **Reliable**
Firstly, there should be a strong feedback loop between the pot and the app so that the pot is purifying air. Secondly, the app itself should be seen as a reliable way to travel to any destination and geocaches should be consistently available.

- **Entertaining**
The gamification feature of the app should serve to engage users so that they feel incited to continue taking greener journeys!

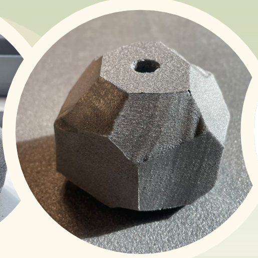
Initial design & CAD prototype

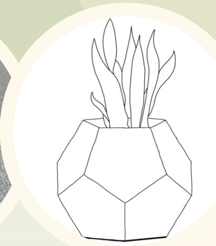
The first stages of our physical product design, including sketches, CAD, stakeholder & user feedback and a low-fi prototype

Final design

Following this, we then prototyped our other favourite designs by laser cutting foam, and found that our **dodecahedron** form resembled **rocks**. By making it regular according to user feedback, we believed that it fulfilled our initial design brief, being **playful, nature-alluding and sleek**. We validated this with **user research**.

"it's a fun design, but would be costly and difficult to manufacture"





Lo-fi prototype

Using our CAD design as a model, we used paper to construct a scaled-down looks-like prototype of our plant pot, and then reproduced a real-scale prototype using card. To answer the **question "what should the screen's location and and size be?"**, we asked 8 individuals what they thought and size, using post-its to indicate potential screens. From feedback, we decided to stick with our initial designs of having it front centre, but slightly decreased the size.

"put it where it would be most obvious- at the front, centre"

"the design would look sleeker if the pots were regular"

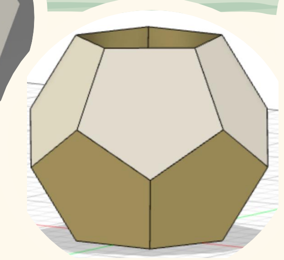
"having a smaller screen means that I need to come closer to the plant pot, which also encourages me to explore its other features"

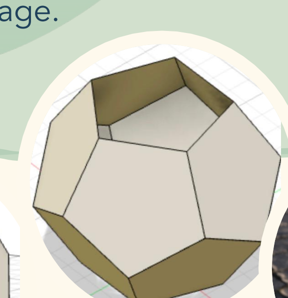
Initial designs

We used our brand essence key words as starting points, aiming for a pot design that was **playful** and **nature-alluding**, yet **sleek**. We started off by collating market research on MIRO, and used our learnings to make sketches. We had decided on a swirly pot design, which we abandoned after **stakeholder** research informed us they thought the design was not **cost-effective** and **challenging to manufacture**.

CAD design

We used our sketches to inform our CAD model. However, it did not feature the LCD screen as we were unsure about its **location** and **size**. We decided to address this in the lo-fi prototyping stage.





Works-like prototype

Making our works-like prototype iteratively according to user feedback

Development

First prototype

We started off with a cardboard, scaled-down prototype of our pot

"Could you change the colour and size so it's more like the real thing?"

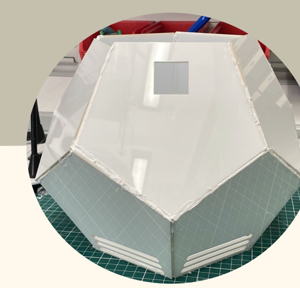
Our response

We decided to make a prototype in white and scale it up

"Could you make it look sleeker and more robust?"

Our final pot

In response to feedback, we decided to use laser cut acrylic sheets instead of cardboard

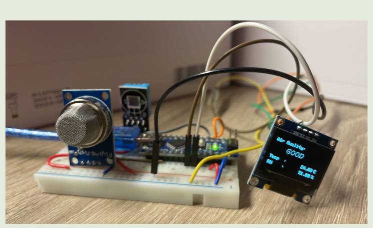


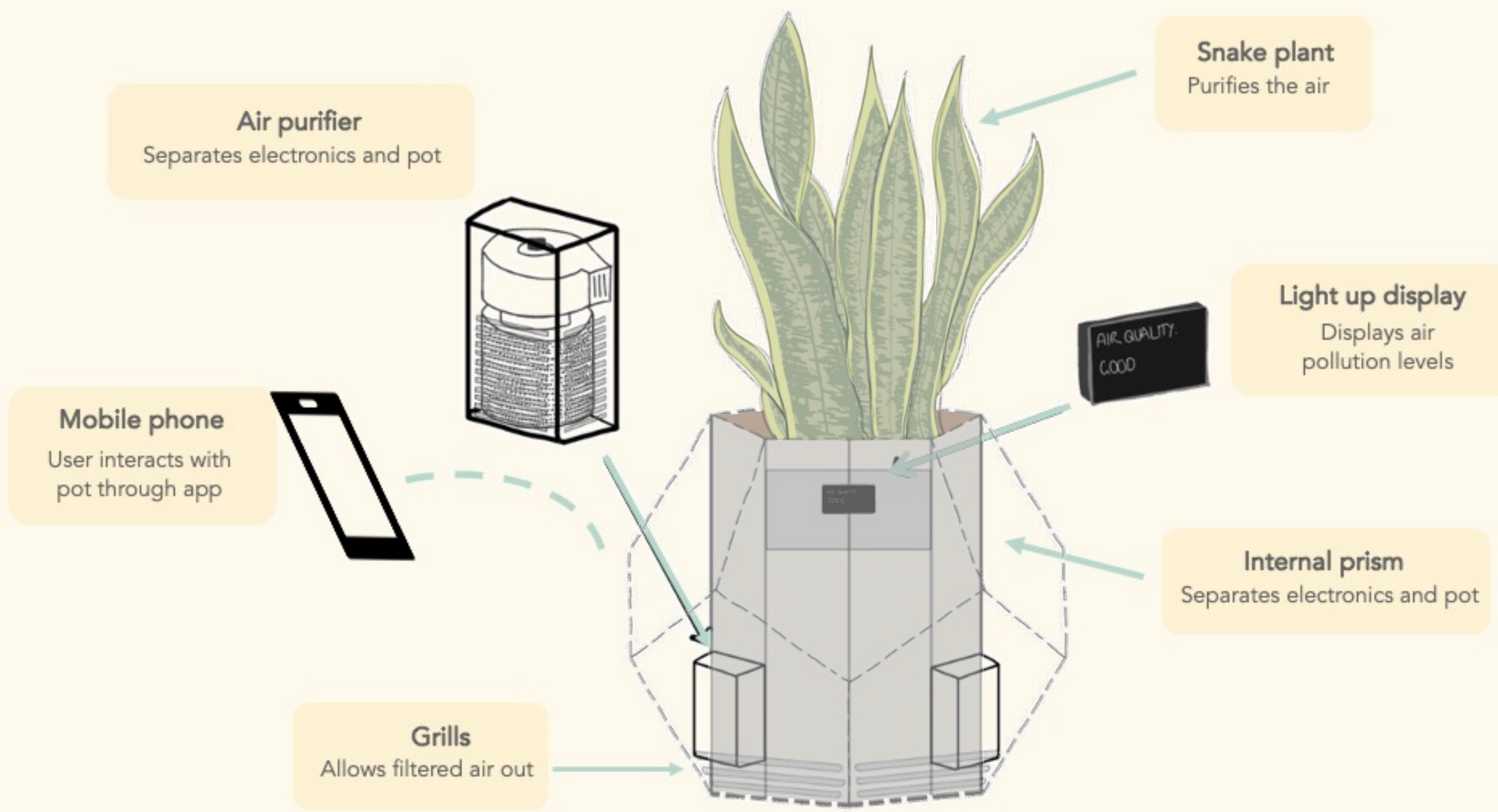
Plant

For our choice of plant, we settled upon Snake plants. They help to filter the air and regulate healthy airflow, and can also convert carbon dioxide into oxygen at night, unlike most other plants. They can remove toxic air pollutants and carcinogens from the air, such as CO_2, benzene and formaldehyde.

Electronics

For our prototype's display feature, we combined electronics with programming on an Arduino nano V3. Components used include an air quality sensor, a temperature and humidity sensor and an OLED display. After receiving user feedback about making the pot more realistic, we decided to add a fan to imitate part of the air purification process.





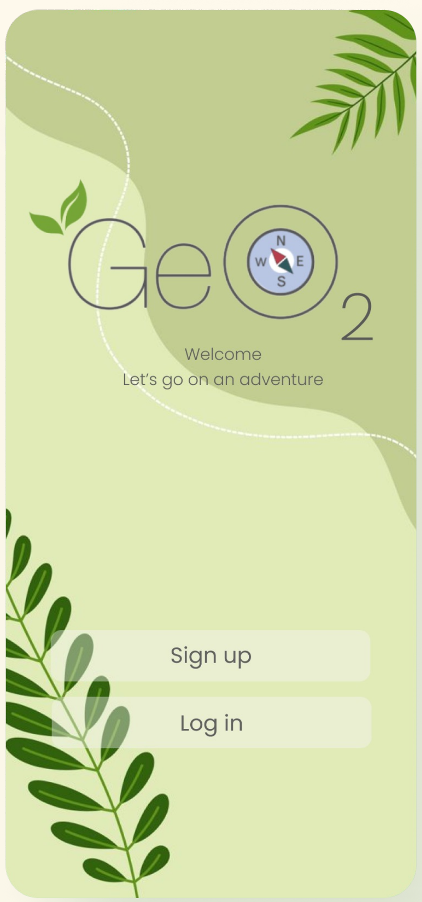
App looks-like prototype

The looks-like UI prototype and feedback for the app which allows users to interact with the air-purifying geocaches

Feedback

We decided to create a look-like prototype for the app, as it is one of the interfaces connecting the geocaches and user. The app maps users' journeys by the path with the least air pollution. Users gain an 'achievement' when they pass geocaches. The geocaches supply air pollution data to the app, which reports this back to the user.

Feedback	Response
"The colours on the first page don't go well together."	Colours changed for readability and accessibility.
"I think there needs to be more incentive to use the app."	Badges, achievements and leaderboard implemented.
"I feel like I might forget to check exposure as its hidden"	Regular notifications reporting exposure

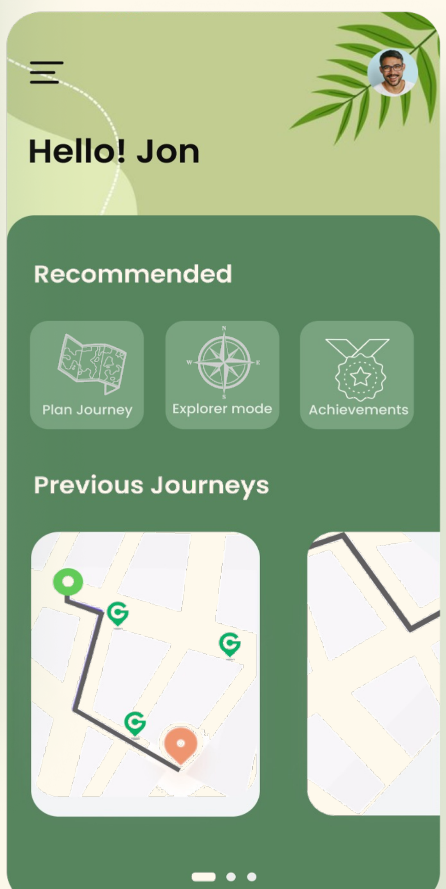


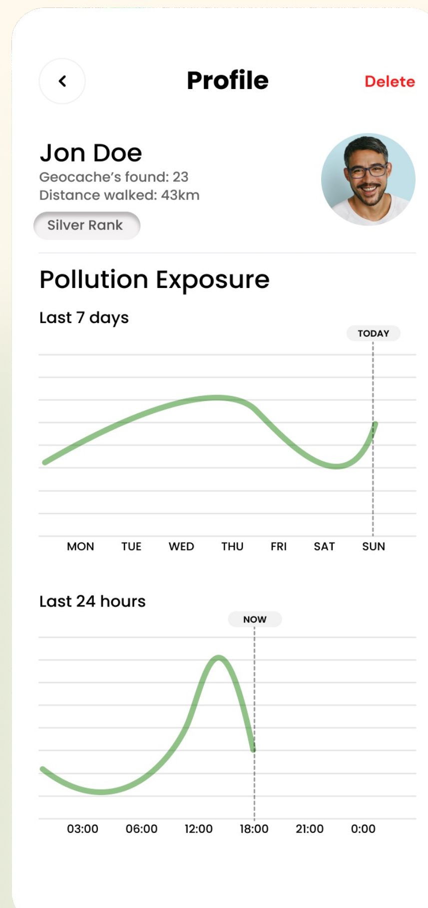
Login page

Users are prompted to log in/ create an account. The nature motifs reflect the values and objectives of the product.

Home page

Here users can choose if they want to plan a journey (which will take them on a less polluted route with geocaches), explore local geocaches, or view achievements.



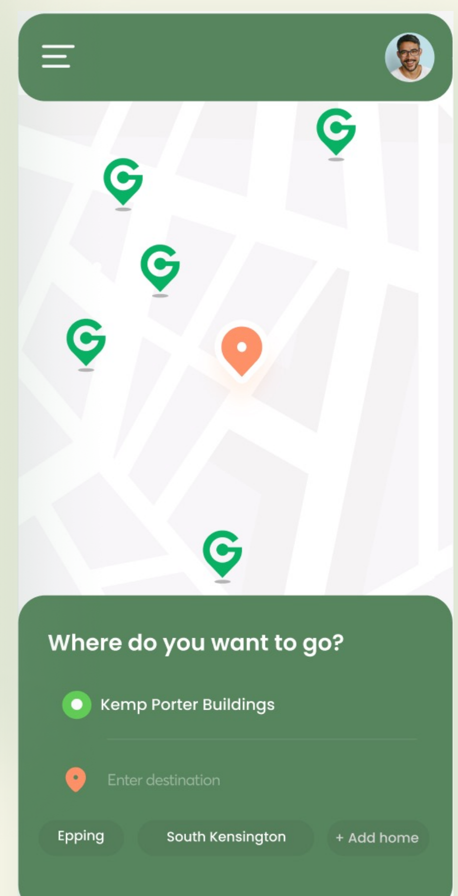


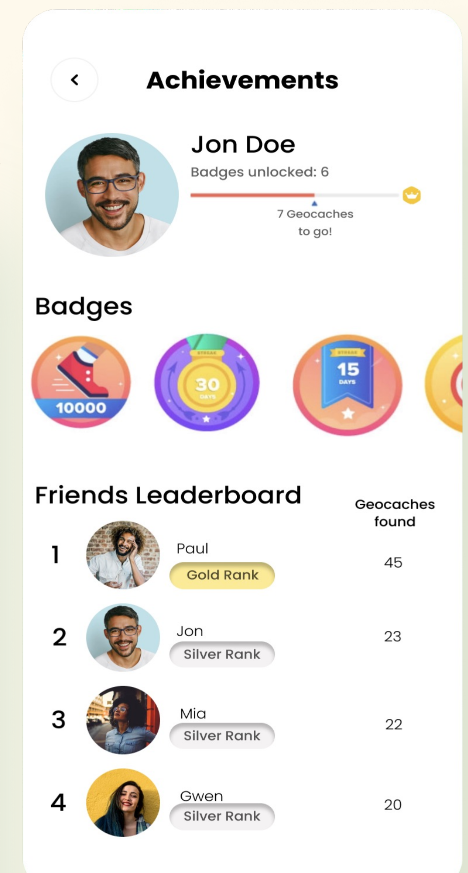
Profile

Clicking on the profile icon in home will show the user their pollution exposure levels, captured and sent via Bluetooth by geocaches they have passed.

Journey mapping

Selecting 'plan journey' on home will open a mapping interface. When entering a destination, the journey will be selected depending on the cleanest route, according to geocaches' data.





Achievements

Selecting 'achievements' from the home page will show the user the amount of geocaches they have passed, as well as achievements and friends' scores, to create incentives to keep going!

Final concept and validation

Our final product and brand, validation and a summary of most adjustments we made during the design process according to feedback

GeO$_2$

Inspiring Conscious Commuting

Ge0$_2$ aims to combat the worsening issue of exposure to air pollution in London through a **dual** approach: by cleaning polluted air and encouraging users to travel through cleaner areas. The plant pot is multifunctional, serving as an **air purifier**, a **geocache**, an **air quality monitor** and a **container** for already air-purifying plants. They are distributed along streets and send their collected data to users' phones via an **app**, which uses this to **notify users** of their exposure, as well as encouraging them to commute by **the path with minimal air pollution**. The **gamification** feature brought by the geocaching concept means that passing a plant pot geocache feels like a reward, and users can **compete** with friends and unlock **achievements**. We believe that by tackling air pollution exposure from two approaches, we can take a significant step towards **a safer and cleaner future**.

Validation

Based on stakeholder's opinions and research:

Users need an incentive to choose slightly longer but cleaner routes
- Gamification with geocaches adds excitement to commuting

Users are most willing to change their behaviour when they can visualize the effects of air pollution.
- Air pollution data is plotted on the app, notifications are sent, and achievements can be won relating to it

Should be accessible to all commuters in London
- Geocaching is an activity that is enjoyed by all ages. Mapping interface is familiar, and amount of text in app is minimized.

Plant pot must be adapted to its multifunctionality in a safe way
- Plant pot and electronics are separated via thick plastic casing.

Adjustments

Summary of adjustments made based on user feedback:

Users had doubts about the effectiveness of the titanium dioxide material, so air purifiers were added inside the pot on the internal prism. Grills added to the pot face to allow filtered air out.

Badges, achievements and leaderboard implemented after users suggested more incentives were needed.

Colours changed in app to improve readability and accessibility.

Abandoned swirly pot design after stakeholders told us it was challenging to manufacture and not cost-effective

Users believed that an irregular pot shape would look unaesthetically appealing, so Design changed to a regular dodecahedron design.

Regular notifications reporting exposure added after users suggested air exposure data was too concealed

Users answered our questions about screen location and size, after which we placed it centre, and slightly decreased the size.

Supplementary pages

References

Page 1: clipart symbols: online webfonts; https://www.onlinewebfonts.com

Page 4: clipart pot and phone symbols: Vector Stock;
www.vectorstock.com/royalty-free-vector

page 5: iceberg clipart: dreamstime; www.dreamstime.com

page 6: pokemon go image: Pokemon; www.Pokemon.com

page 6 AIRY pot image: TechAcute, www.TechAcute.com

Layout and formatting

Header- Aharoni bold (54)

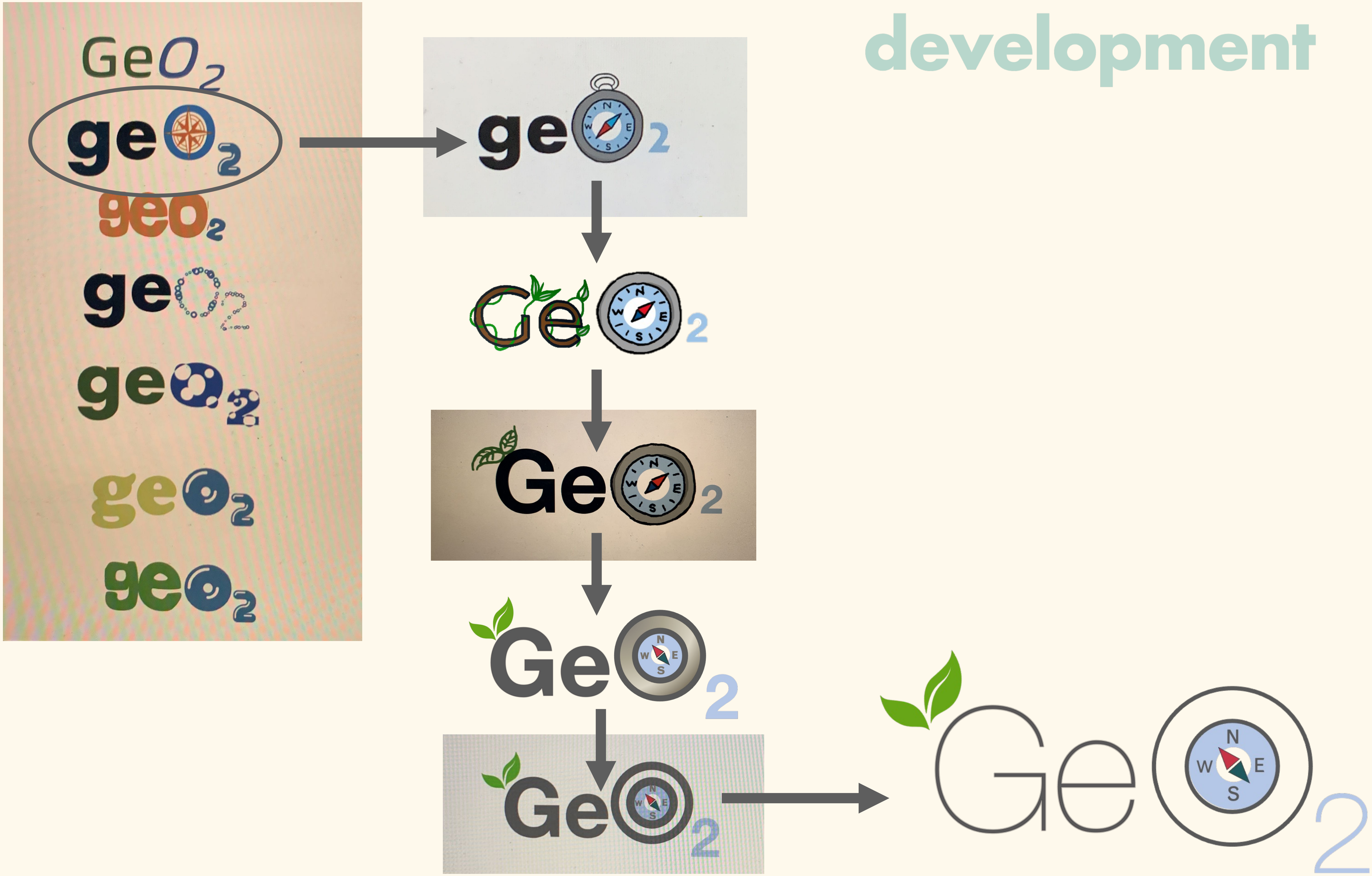
Sub-heading: avenir, bold (36)

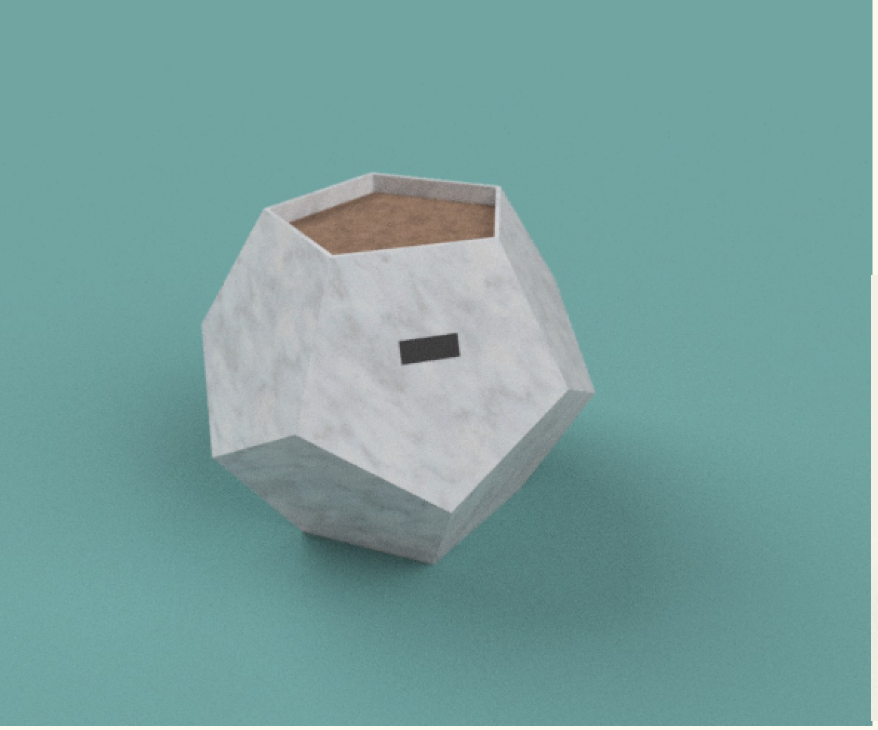
Paragraph- Century Gothic (12)

Paragraph emphasis- Avenir (18)

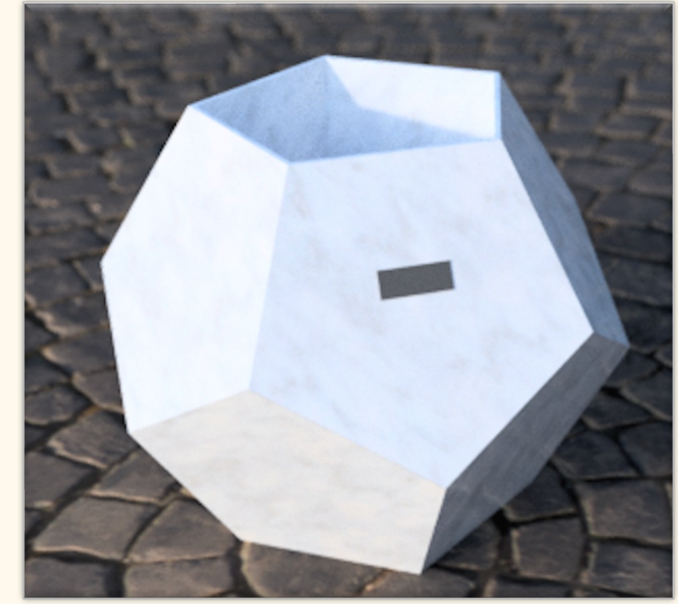
Presentation layout guidelines

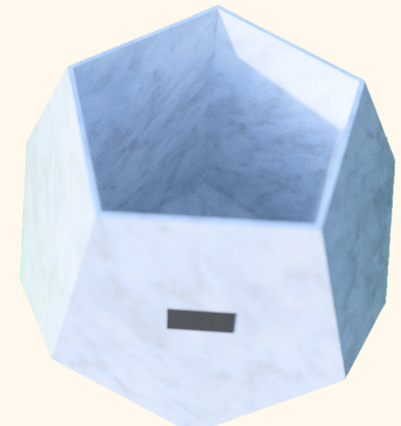
Logo development



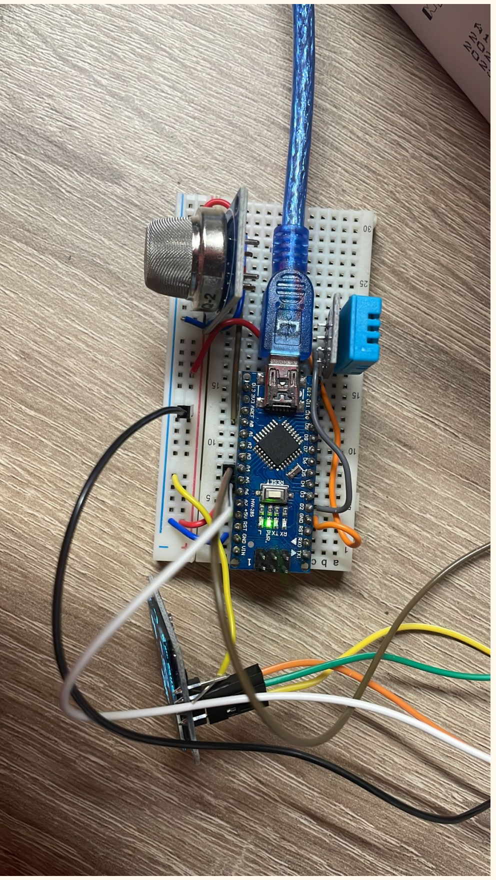


CAD model

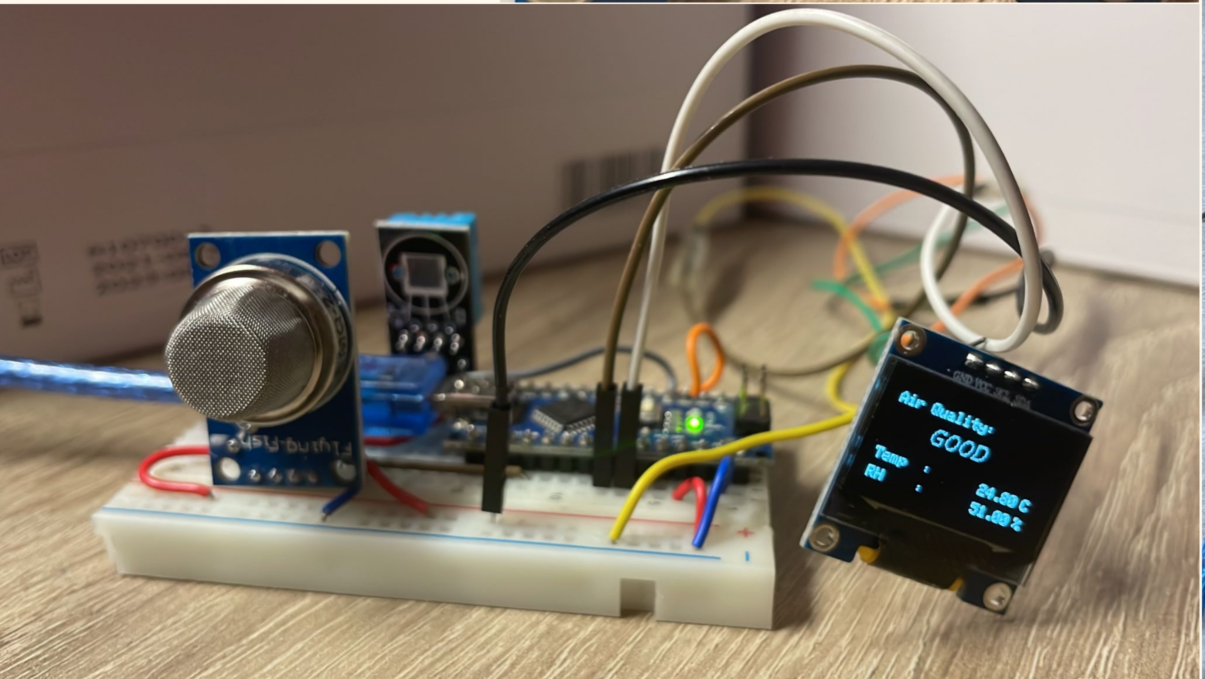


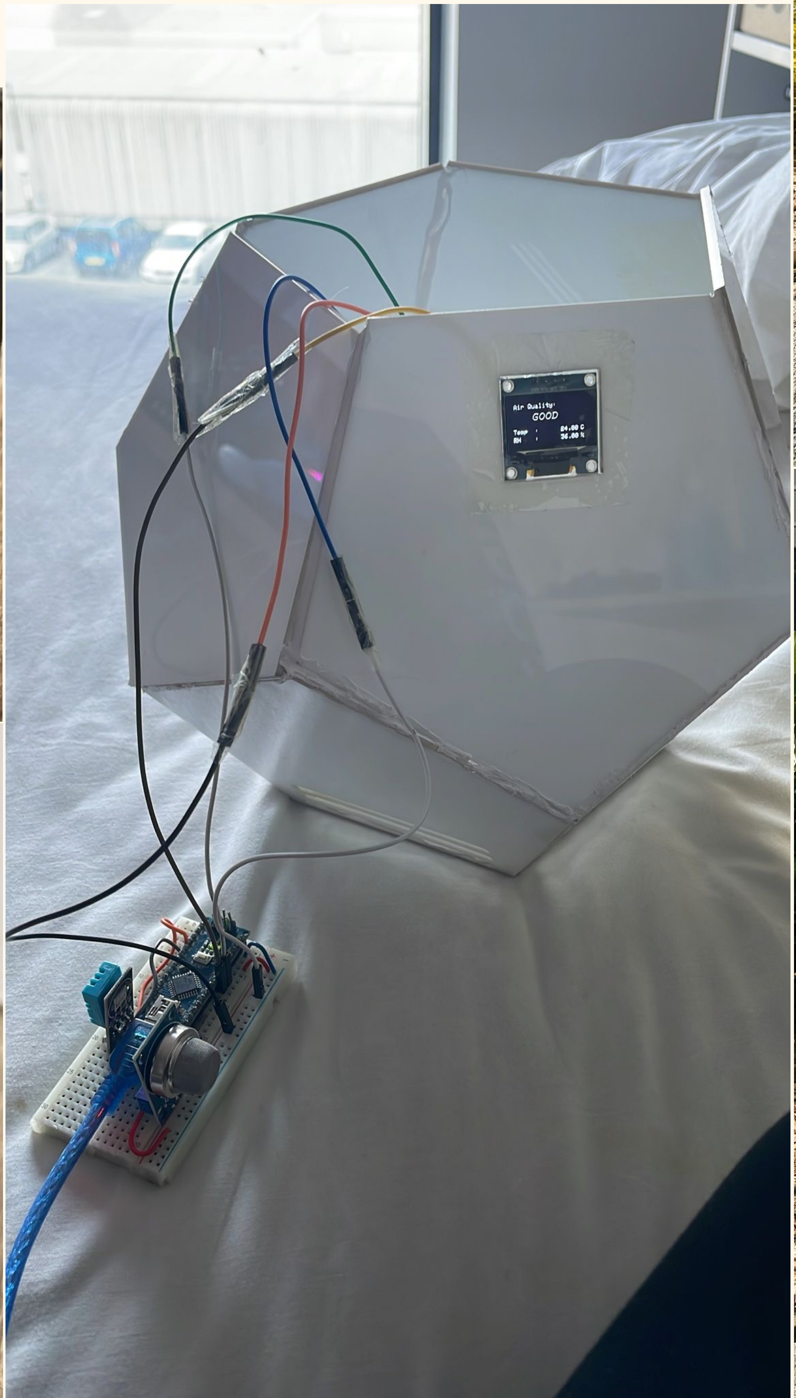


Prototype




GND VCC SCL SDA
Air Quality:
GOOD
Temp : 25.30 C
RH : 50.00 %


Air Quality:
GOOD



Arduino code

Plant species

Snake plant - Sansevieria trifasciata

The snake plant is an evergreen plant which means our flowerpots will flourish all year round. Snake plants are very easy to care for and require very little water to survive. Snake plants are suitable for all continents. This means that our pots can survive in more sheltered areas as well as on the streets.

Snake plants help to filter the air and regulate healthy airflow. What's special about this plant is that it's one of the few plants that can convert carbon dioxide into oxygen at night.

Snake plants can remove the following toxic air pollutants and carcinogens such as:

- CO_2
- benzene
- formaldehyde
- xylene
- trichloroethylene
- toluene

Meeting minutes

https://imperiallondon-my.sharepoint.com/:w:/g/personal/ew821_ic_ac_uk/ESadD9_KUE5OnjPwFcOBX90BqERw_F-rSDdNZMukY2xOyQ?e=9n0DM0